HUMAN OVERSIGHT SYSTEMS LAB
Stevens Institute of Technology

Oversight That Degrades

Architectural Failure Modes in Human-LLM Control Structures

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Human review can collapse from active validation into procedural approval, while appearing intact. The most dangerous failure modes are the ones that look like normal operations.

TOP RPN
504
Authority Drift
DEGRADED CENTAUR

THE PROBLEM

AI safety frameworks assume human oversight functions as a reliable control mechanism, specifying that oversight must be present without specifying the control architecture through which it operates, or the conditions under which that architecture degrades.

01 · A control-structure taxonomy mapping human-LLM collaboration patterns to safety-relevant failure modes.

02 · Qualitative observational evidence consistent across two contexts with opposing governance regimes.

03 · A structured failure-mode assessment showing detection difficulty, not severity, as the dominant safety concern.

THREE-PHASE DEGRADATION PATTERN

PHASE 01
01
Active validation
Reviewers engage substantively, errors are caught, and the checkpoint is load-bearing.

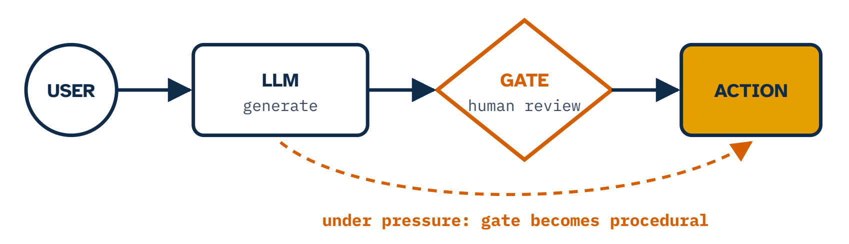
PHASE 02
02
Procedural validation
Reviews occur and approvals are recorded, yet functional engagement has receded; the governance record is indistinguishable from Phase 1.

PHASE 03
03
Automated approval
The checkpoint is a formality and the human role is acknowledge, not evaluate. Organizational records in Phase 3 typically resemble those in Phase 1.

No recent failures, systematically mistaken for system stability.

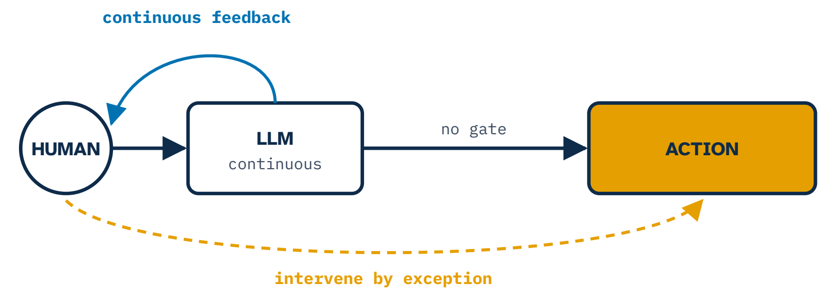
TWO ARCHITECTURES, TWO FAILURE SIGNATURES

Centaur *Gated validation control*



Failure looks like approval.

Cyborg *Supervisory control*



Failure looks like drift.

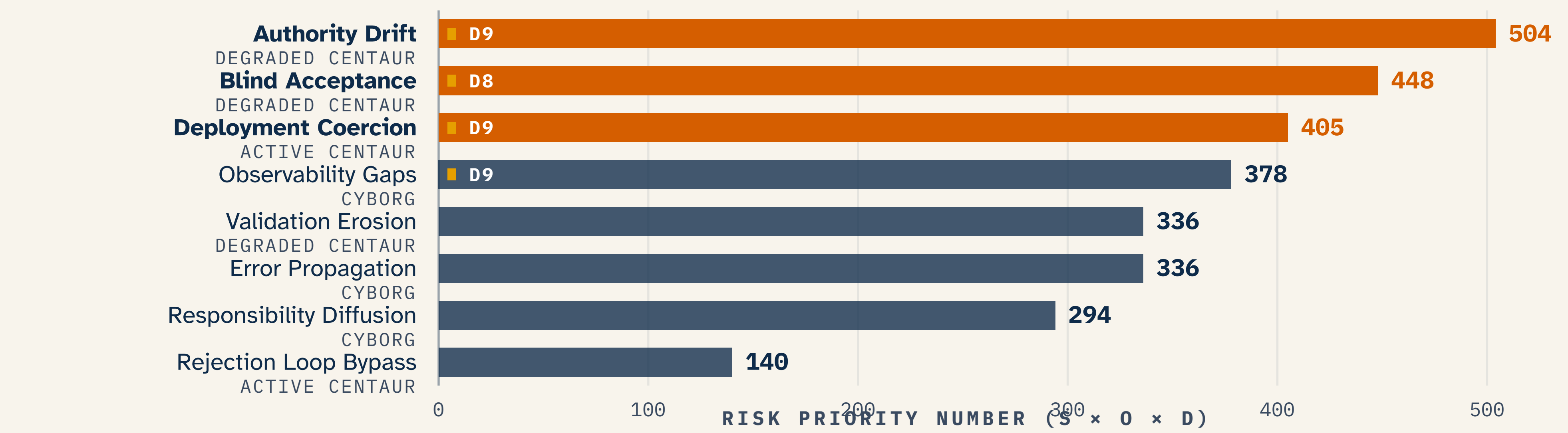
SAME GOVERNANCE DIMENSION · TWO ARCHITECTURAL ANSWERS

CENTAUR		CYBORG
Concentrated at one checkpoint	AUTHORITY	Distributed across the loop
Discrete pre-execution gate	VALIDATION	Continuous in-flight monitoring
Per-output approval record	AUDITABILITY	Process-level activity record
Pre-execution approval	INTERVENTION	Exception-based override
Discrete output review	OBSERVABILITY	Ongoing state monitoring
Silent collapse at gate	FAILURE SIGNATURE	Diffuse drift across loop

FMEA · FAILURE MODES RANKED BY RISK

S Severity · how bad it is if it happens **O Occurrence** · how often we expect it **D Detection** · how hard it is to notice RPN = S × O × D (each 1–10)

48% of total risk concentrates in the **top three modes** — all of which score D ≥ 8 (detection-hard). Detection, not severity, is the binding constraint.



TOP-3 RPN OTHER MODES D ≥ 8 DETECTION-HARD

Top three modes account for the majority of total risk and all score D ≥ 8 — meaning detection, not severity, is the binding constraint.

Reactive governance may systematically miss the highest-risk failure modes, because those failure modes are defined by their undetectability.

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